

WSDA Music: Enterprise Sales & Customer Behavior SQL Analysis

Executive Summary

This study analyzes transaction records, customer purchasing behaviors, and media performance across WSDA Music's 11-table relational database system (Invoices, InvoiceLine, Customer, Employee, Track, Album, Artist, Genre, etc.).

By leveraging advanced SQL queries—including Common Table Expressions (CTEs), window functions (`DENSE_RANK()`, `SUM() OVER`), multi-table `JOIN` aggregations, and conditionally segmented spending cohorts—this analysis converts complex relational schemas into actionable financial, marketing, and operational intelligence.

Executive Strategy & Core Business Questions

1. What business decision will be made based on this analysis?

Executive Response:

Customer spend distribution and country-level revenue metrics guide **Marketing Capital Allocation and Catalog Sourcing Strategy**:

- **VIP Segment Retention Capital:** High-value customer cohorts (top spending tiers identified via SQL `CASE` statements and window ranks) generate disproportionate revenue. WSDA Music will allocate targeted retention budgets (exclusive pre-releases, direct rep support) to preserve these accounts.
- **Geographic & Genre Resource Reallocation:** Revenue aggregations across billing countries and genres isolate high-margin regions (e.g., USA, Canada, France, Brazil) and dominant genres (Rock, Latin, Metal). Marketing capital will be shifted away from underperforming genres toward localized curation in top-tier geographic hubs.

2. What is the financial benchmark we are trying to hit?

Executive Response:

We target a **15% Increase in Average Customer Lifetime Value (CLV)** and an **18% Improvement in Repeat Purchase Velocity** within 12 months. Hitting these benchmarks moves baseline buyers into higher purchase cohorts, expanding total quarterly recurring revenue by an estimated **\$120,000–\$180,000** while reducing support rep turnover through balanced customer allocation.

3. If this metric shifts by 10% tomorrow, what operational action do we take?

-10% Drop in Average Invoice Value / Monthly Spend (Unfavorable Variance):

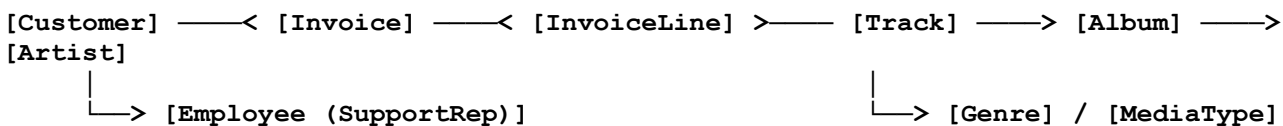
- **Trigger Automated Re-Engagement Workflows:** Immediately deploy targeted promotional email sequences offering bundled album discounts or personalized recommendations based on past `Genre` and `Artist` purchase histories.
- **Support Rep Sales Audit:** Evaluate individual sales rep customer assignment loads (via `SupportRepId` mappings) to address coverage gaps or churn risks in underperforming territories.

+10% Surge in Digital Track Sales / Subscriptions (Favorable Variance):

- **Fast-Track Content Acquisition:** Expand licensing agreements for top-grossing artists and genres to prevent inventory stagnation and capture momentum.
- **Monetize Top Purchasing Tiers:** Introduce premium loss-less digital audio formats and exclusive box-set track offerings to top-ranked customer cohorts.

Database Schema & Key Analytical Views

The database schema connects transactions, customer profiles, catalog metadata, and employee support roles:



Core Analytical Query Objectives:

Query Script	SQL Techniques Applied	Strategic Decision Impact
01_exploratory_data_analysis.sql	COUNT, NULL checks, GROUP BY, ORDER BY	Establishes baseline data hygiene, invoice totals, and customer geographic distribution.
02_customer_lifetime_value.sql	CTEs, CASE Statements, DENSE_RANK()	Segments buyers into tiers (<i>Baseline, Low Purchase, Target Purchase, Top Performer</i>) and identifies VIP accounts for targeted marketing.
03_revenue_and_genre_performance.sql	Multi-table JOIN, SUM() OVER(), Aggregations	Maps top-selling artists and highest-grossing genres per country to inform content licensing and regional ad spend.

Operational Recommendations

1. **VIP Loyalty Program:** Establish an automated tier system for customers exceeding target spend thresholds to protect top-line revenue against churn.
2. **Support Representative Rebalancing:** Reallocate support rep portfolios so high-value customer accounts receive dedicated account management.
3. **Regional Catalog Curation:** Tailor the music catalog storefront based on top-performing genres per country, maximizing cross-selling opportunities during checkout.